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CS3821 Full Unit Project - Interim Retrospective

An Intelligent Recommendation System to Promote Reading Habits

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Chapter 1: Introduction

This report provides a detailed reflection on Term 1, comparing the intended project plans with what was actually achieved, and outlines the planned work for Term 2.

Chapter 2: Term 1 Reflection

W/C	What I had planned to do	What was actually achieved	Comments
13.10.25	• Finish initial Figma Designs • Research architectural styles and design patterns and write up justifications for the choices made • Research best database storage practices to handle book metadata and write up justifications • As a user, I want to view books on the web app so that I can read	• End-to-end connection achieved with API connections • Books fetched from Google Books API and displayed on front-end • Began setting up my interim report • Structured the layout of interim report • Added user stories and high-fidelity figma prototypes to software engineering section • First draft of State of Art Web Development	I struggled to set up the initial end-to-end connection so that I could achieve my first high priority user story which is why I had less time to work on the report. Also, the report was very overwhelming initially and the requirements were unclear to me as well. I thought I needed to quickly decide on my architectural styles and design patterns rather than take time to explore all options and write up all my research in the background theory section. Next term I will ensure to fully clarify all aspects of the report with my supervisor prior to writing to mitigate the confusion.
20.10.25	• As a user, I want to sign up securely using email or through TikTok/Netflix OAuth so that I can start personalising my account. • Write up justifications for using OAuth as a secure practice. • As a user, I want the web app to be visually similar to entertainment apps I am familiar with so that I can engage with the website easily. • As a user, I want books to be displayed in organised categories so that I can find books I will enjoy easily. • As a user, I want a navigation bar with "Home," "Genre," and "My Library" options on the left, and "Search," "Profile," and "Notifications" on the right so that I can easily navigate to different sections of the web app.	• Navbar implemented • BookTest, BookControllerTest (still needs fixing) • SecurityConfigTest, SecurityIntegrationTest classes implemented	I had a written assignment due for one of my modules which significantly impacted my FYP progress, so I only managed to achieve the Navbar. Also, I didn't consider the fact that integrating other platforms into my log-in would be too complex at this stage - I later decided to use JWT authentication. I was quite idealistic this term with my desired approaches which resulted in me having to make changes in my plans. So, my term 2 plans are much more careful and realistic to mitigate this issue from arising again.
27.10.25	• As a user, I want to see a personalised "For You" feed of book cards so that I can discover books tailored to my preferences. • Write up justifications for chosen machine learning algorithms and any challenges I faced.	• CI/CD pipeline configured • Fixed broken test classes • Added javadoc and checkstyle	After the previous week of slowed down progress, I had to try and catch up with what I missed. However, I realised my IDE had a lot of configuration issues with javadoc and checkstyle which took a very long time to fix. I also struggled a lot with the pipeline configuration. In terms of the report, at this point I started to realise my planning was not directly aligned with the requirements due

			to misunderstanding so I prioritised product progression. Also, I was still not experienced enough with my machine learning knowledge to start implementing an algorithm from scratch at this stage. Now that I have completed my machine learning module and have carried out more research, next term this should no longer feel tedious.
03.11.25	- As a user, I want to see a personalised “For You” feed of book cards so that I can discover books tailored to my preferences. - As a user, I want to swipe or click arrows to move horizontally through book cards so that it feels intuitive and dynamic, like Netflix. - As a user, I want book cards to visually enlarge on hover and display an enticing quote so that I’m immediately drawn to explore further.	- IntelliJ IDE issues solved including checkstyle config mismatches - Implemented authentication page with end-to-end functionality and JWT - First draft of architectural design patterns and paradigms - Second mandatory meeting with supervisor: Discussed current interim and product status - Fixed book fetch and user accounts deleting bugs - Added Swiper module to allow navigating across book rows (left and right) with fade effect - Redraft of architectural design patterns and paradigms section started - Books that were originally fetched were quite old and academic, now they are popular thrillers and romance novels	I really started to pick it up this week and managed to implement JWT authentication which I found the most suitable due to Spring Boot’s support. I also managed to implement the arrow swipers and fetch books based on category. After my second supervisor meeting it was much clearer what the report requirements were so I was able to start working on it properly. As mentioned, next term I want to clarify all aspects of the report as soon as possible so that this does not happen at such a late stage.
10.11.25	- As a user, I want to start reading within the app using an in-built timer so that I can track my active reading time. - As a user, I want the app to record the last page I reached when I stop the timer so that I can easily continue next time. - As a user, I want to earn points or streak rewards for consistent reading so that I stay motivated. Write up justifications and challenges faced for the gamification element.	- Added search bar functionality — currently performs local filtering to avoid repeated API calls - Searching also normalises input - Added a fade effect when scrolling through the cards - Genre tab is being worked on...currently displays some genres that a user can select and all books with that genre are displayed to the user with an animation - What are architectural paradigms and design patterns section... - Architectural paradigms: - Object-oriented complete - Design patterns: - Creational patterns: Singleton, Abstract Factory, Builder - Structural: Adapter, Facade, Bridge	This is the week where I began to struggle following my project plan as it had been a couple of weeks where I could not meet the targets and my report planning was not effective. So, I decided to work on some lower priority user stories as I had overworked myself the week before. I also wanted to focus on the report as well so I needed to make sure my implementation requirements for the week were achievable. If this occurs next term, I will make sure to push the more challenging tasks to less busy weeks and vice versa.
17.11.25	- As a user, I want to see my current reading streak and daily goal progress displayed near the navigation bar so that I can stay encouraged. - As a user, I want to view my total points and achievements so that I can measure my long-term progress. - As a user, I want to add friends and compare progress so that reading feels social and rewarding.	- This week I worked on my report as I was falling behind - Completed Web development Architectural paradigms and Applicable Design patterns section - Wrote introduction for state of the art web development and completed subsection on React	Again, at this stage I stopped following the plan as it became very unrealistic and tedious. My other modules were also getting much harder to balance with FYP as well as the fact that I still wanted to enjoy my social time at uni. Also, I noticed my report progress was very slow since I prioritised perfectionism, so I decided to prioritise getting sections completed quicker, even if they are less thorough, and

			then focus on polishing and expanding if I have time left. As the deadline approached, I was forced to do this under the pressure. Next term, I will do this first and then perfect my work in dedicated quality checks.
24.11.25	- As a user, I want to comment, like, and add books to lists so that I can interact with other readers. - As a user, I want spoiler comments and spoiler review videos to remain locked until I mark the book as completed so that my experience isn't spoiled and I feel rewarded for finishing. - As a new user, I want to swipe right and left through book cards when I first log in to indicate what I like so that my recommendations are accurate from the start. - As a new user, I want to answer quick onboarding questions (e.g., "When was the last time you read?", "What movies do you enjoy?") so that the app can infer my literary preferences.	- This week I focused on progressing key development tasks, particularly around the My Library feature. - Implemented backend support for adding books to a user's My Library, including creating a new <code>user_library</code> table and updating the User model. - Added a new <code>LibraryController</code> with token-based authentication and updated Spring Security to resolve a 403 error. - Wrote and updated JUnit tests to reflect the new secure logic. - On the frontend, made the Add to My Library button functional and improved hover behaviour on book cards.	Returning to development this week felt productive after focusing heavily on the report previously. Working across backend, security, and frontend showed the complexity of full-stack features so I aim to maintain this momentum without over-complicating implementations.
01.12.25	- Clean up code, debug and ensure all tests are working correctly - Update Figma designs to be clickable with UI flow and main interactions - Finish up missing sections from the report	- Prepared for supervisor meeting by completing as much of the report as possible - I completed my first drafts of all sections (including bullet points to make sure I had at least something each section) - I only left the literature review blank I was still unsure how to do it - Final supervisor meeting where all misunderstandings with the report were clarified, also product progress was evaluated	By this stage I was very mentally drained and it was becoming much harder to continue working on the project as it reached the end of the term and I wanted to spend time with friends before the holiday. However, I still tried my best and my confidence with the report increased even though the sections weren't final, they were no longer empty. As mentioned above, I will make sure to do rough drafts and move on to the next sections and at a later stage return and perfect my work.
08.12.25	- Prepare for end-of-term presentation - Start planning term 2 goals	- Prepared for and completed the interim presentation - I decided to get the 5-day extension as I had a machine learning assignment which I had to prioritise so very minimal progress on the report and development	The presentation was nerve-wracking but enjoyable, and it was interesting to see what my supervisor group had produced this term. It was also comforting to know that I wasn't the only one who found it very challenging. After the presentation I had to focus on my machine learning assignment which was also very difficult so I needed to take the extension for interim.
15.12.25	N/A	- Completed the report, retrospective and demo video for the interim submission - Cleaned up existing code and submitted zip file	I do want to continue working on my project over the holiday as I feel I can do a lot more with the product but I struggled a lot with the time management and multitasking. This was definitely the most challenging academic piece of work I have had to do but I have grown a lot from it and I feel much better prepared for next term as I know what worked and what didn't.

Chapter 3: Term 2 Project Timeline

Below is the project timeline for term 2. I have based the allocated tasks on the 10-hour minimum target each week as well as the difficulty. There is a possibility I will achieve some tasks over the holiday but I am going to assume I don't manage to complete anything given that I have two big assignments to work on already. If I do manage to complete some tasks over the holiday I will use that time for further quality checks and optimisations. Again, these are educated estimations and inaccuracy is inevitable, but after experiencing term 1 it makes it easier to plan better.

Also, with awareness that studying for the end of year exams will start to creep up, I have pushed the more complex implementations to Jan-Feb and the simpler to March and April.

Product implementation

Project Task	Est. Time (Days)	Starting Date	Hard Deadline Date
Risks and mitigations	Start reading timer and save pages read to database	2-3 days	19 Jan
23 Jan	Reading progress and continue reading feature	3-4 days	22 Jan
28 Jan	Points system based on pages read	2 days	29 Jan
2 Feb	Reading streak feature on navbar	2-3 days	2 Feb
6 Feb	Reading streak calendar - open/closed book icons displaying days read/not read	4 days	6 Feb
12 Feb	Design hybrid recommendation approach (content based + collaborative)	4-5 days	20 Jan
14 Feb	Research intelligent semantic searching and design algorithm	7 days	27 Jan
21 Feb	Implement basic recommendation pipeline (data features, scoring logic)	5-6 days	17 Feb
14 Mar	Implement basic prompt-based search algorithm	4 days	17 Feb
14 Mar	Adding friends feature	3 days	12 Feb
18 Feb	View friends points feature	1-2 days	19 Feb
23 Feb	Leaderboard feature - comparison based on number of pages read weekly	4 days	24 Feb
27 Feb	Achievement badges with hover feature to display description of badge - initial badges based on number of books read	2-3 days	2 Mar
6 Mar	Analysis board feature - categories of books read	3-4 days	9 Mar
13 Mar	Add comments on books feature	2 days	16 Mar
19 Mar	Spoiler and non-spoiler comments - spoilers locked	4 days	20 Mar
25 Mar	Research text analysis algorithms for identifying spoiler comments	6 days	26 Mar
1 Apr	Points system optimised based on factors like difficulty, first-time reader award, streak	3-4 days	18 Mar
28 Mar	Achievement badges optimised with more variety of badges	2-3 days	18 Mar
28 Mar	Optimise recommendation algorithm	2-3 days	24 Mar
4 Apr	Optimise search algorithm	2-3 days	24 Mar

4 Apr

N/A

N/A

N/A

Report writing

Project Task	Starting Week	Hard Deadline Week
Background Theory: Machine Learning Algorithms for Recommendations Improvement	1	3
Background Theory: Machine Learning for Intelligent Searching	2	4
Background Theory: Features to combat Doomscrolling Improvement	3	4
Technical Decision Making	5	8
Critical Analysis and Discussion	6	9
Software Engineering: Research	7	8
Software Engineering: Design	8	10
Software Engineering: Implementation	9	11
Software Engineering: Testing	10	11
Software Engineering: Deployment	9	10
Final quality check	10	11

The report planning is intentionally less detailed than the product implementation plan, as the writing process is influenced by several external factors. These include the workload and intensity of other modules in a given week, the clarity of individual report sections at the time of writing, and the need to occasionally wait for supervisor feedback before progressing. As a result, it is difficult to produce a concrete plan for the report at this stage, and the outline provided should be viewed as a flexible guide rather than a fixed schedule.

Chapter 4: Planned risk mitigations compared with actual mitigations implemented

Building a full stack project is a challenge, and with that challenge comes risk. Below I will discuss general and project-specific risks that are inevitable with a project of this scale, as well as mitigations I will carry out if and when they occur.

1. Poor time management

- “To mitigate this risk, I will organise my tasks based on priority and work on those first each week. I will also regularly update my timeline to rebalance the tasks and work on completing as many high-priority tasks as possible by the end of the term. As mentioned in the previous section, some weeks have lighter workloads so rebalances should not be a huge problem.”
- I organised my tasks based on priority and focused on implementing a minimum viable product (viewing books, seamless netflix-style experience, navbar and searching). However, a lot of my high priority tasks seemed too tedious with module balancing and report writing in parallel. So, on deadline-heavy weeks I chose to work on the tasks with lower priority that were simpler and quicker to implement so that I still made some

progress. In terms of the timeline having lighter weeks, this was helpful as it meant I could rebalance tasks effectively and still achieve as much as possible.

2. Blockers hindering development

- “To mitigate this risk, I will ensure my code follows best practices in coding quality. This includes regular committing, testing and pushing, as well as clean structure and documentation so that problems are much less likely to arise. If they do, however, the debugging process will be much easier to manage.”
- I used Maven tools such as Javadoc, Checkstyle, and JaCoCo throughout development. I also made consistent use of GitLab issues and ensured that commit messages were clear and descriptive. This made it much easier to track changes, understand previous versions of the code, and identify when and where issues were introduced. As a result, I found that debugging became less time-consuming and more structured.

3. Poor computational efficiency

- “To mitigate this risk, I will conduct further research on recommendation algorithms used by social media applications that worked best to ensure efficient code from the start.”
- I carried out significant research on recommendation algorithms that are suitable for my project. Although I did not manage to implement the algorithms, I feel much more prepared for next term when I do.

4. Hardware failure

- “To mitigate this risk, I will ensure my code is regularly committed using Git (a version control system) to a remote repository (GitLab) to prevent reliance on local data. I will also work on my report on Typst, a cloud-based document editor.”
- This mitigation was effective. Commits were made regularly and I consistently pushed them to the remote GitLab repository, mainly on a feature branch, with merges into the main branch only occurring once the code was clean and functional. This reduced reliance on local data and provided a reliable backup of the project. Also, the report was written entirely using Typst, which meant I could work from different devices and locations at any time.

5. Poor user engagement

- “To mitigate this risk, I will regularly confide with the target market (14-16 year olds) and receive feedback to ensure high engagement.”
- I received some user feedback from people I knew of the target market, and this helped me understand what requires optimisation/ommitting. I intend to request further user feedback when I implement more features in second term.

6. User data exposure or poor input validation

- “To mitigate this risk, I will implement prepared statements, validation, and encryption, as well as research best methods for maintaining security in the system.”
- I used Spring Boot Security with JWT-based authentication and secure password hashing, ensuring that sensitive user data was protected and access to endpoints was restricted to authorised users. While the application relies on Spring Data JPA, which reduces the risk of SQL injection through parameterised queries, I intend to add further input validation and review database access patterns in the future.